

Problem Sheet 5 Solutions

1.

(a) The amplifier has a closed-loop gain

$$G = 1 + \frac{10^6}{3 \times 10^3} = \frac{1003}{3}$$

$$G_{dB} = 20 \log G \approx 50.$$

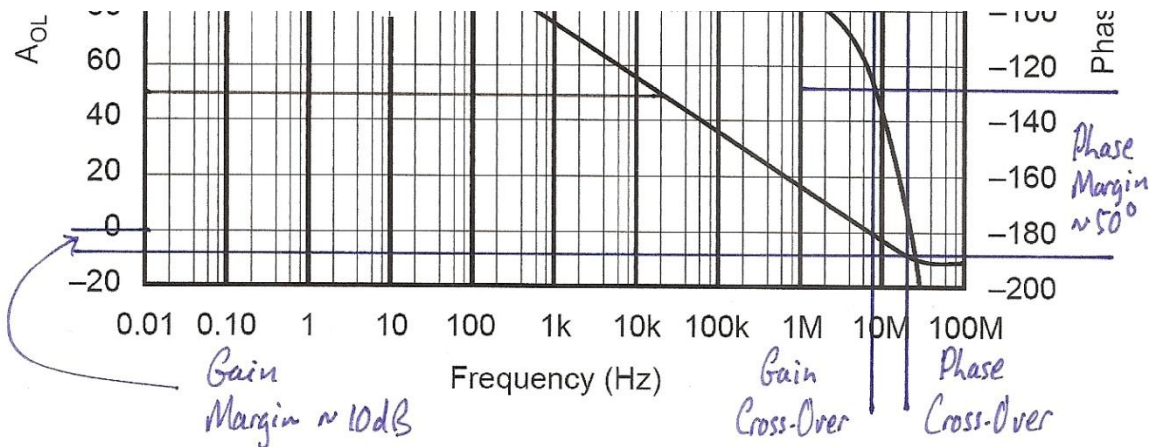
Sketching a horizontal line at 50 dB onto the figure, we estimate this intersects the open-loop gain curve at around 20 kHz.

Note also that we can estimate the gain-bandwidth product for this op-amp to be about 7×10^6 (gain is 0 dB at 7 MHz). Hence, we would expect the closed-loop bandwidth to be

$$7 \times 10^6 \times \frac{1003}{3} \approx 20 \text{ kHz}.$$

(b) The closed-loop roll-off must be the same as for open-loop. We know this should be -20 dB/decade and we can verify this from the figure.

(c) From the figure below we can make an estimation of the phase and gain margins. The phase margin is about 50° and the gain margin is about 10 dB.



(d) For a given op-amp, the voltage-follower (buffer) is the most sensitive from a stability point of view since the loop gain βA is a maximised in this configuration, as $\beta = 1$. Nevertheless, stability will be ensured if we have positive gain-margin and positive phase-margin, i.e. the phase stays below 180° while the gain is greater than 1. We predict that the voltage-follower will be stable, and we would describe the OPA227 as inherently 'unity-gain stable'. Note that the non-inverting amplifier configuration of part (a) would have a bigger margin of stability since $\beta < 1$.

Instrumentation

2.

(a) The total number of possible digital values is $\frac{1}{0.125} = 8$ (the dynamic range of the converter). Therefore, we will need 7 comparators.

(b) To represent the digital numbers 0 decimal to 7 decimal (111 binary) requires 3 digital bits.

(c) See diagram below.

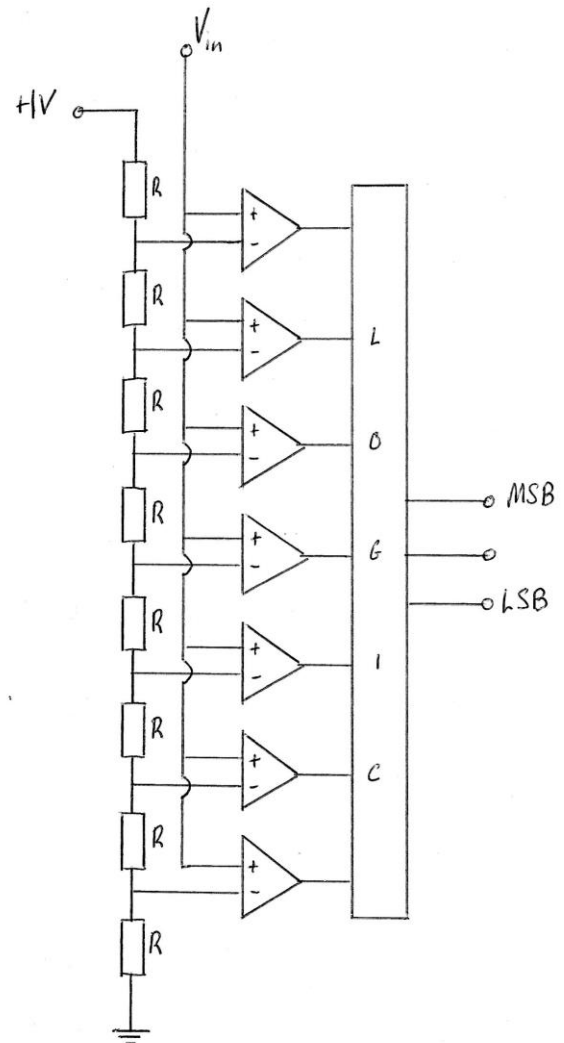
(d) Refer to table below which numbers the op-amps in the figure from bottom to top.

V_{in}	7	6	5	4	3	2	1	binary	decimal
>875 mV	T	T	T	T	T	T	T	111	7
750...875 mV	F	T	T	T	T	T	T	110	6
625...750 mV	F	F	T	T	T	T	T	101	5
500...625 mV	F	F	F	T	T	T	T	100	4
375...500 mV	F	F	F	F	T	T	T	011	3
250...375 mV	F	F	F	F	F	T	T	010	2
125...250 mV	F	F	F	F	F	F	T	001	1
<125 mV	F	F	F	F	F	F	F	000	0

(e) An analogue input of 1.2 V results in positive saturation of the ADC. The output is b111. An analogue input of -0.125 V results in negative saturation of the ADC. The output is zero.

With an input of 0.5 V comparator number 4 will be on the threshold between positive and negative output. Since the open-loop gain of the comparator is 10^6 or more, a few μV difference at the input is sufficient to swing from +ve to -ve saturation. Given that there will always be some small variation and noise on the input signal, the output could wander between adjacent digital values. Practical ADC designs will implement a small amount of hysteresis to prevent this.

(f) The purpose of the resistor chain is simply to bias the op-amp inverting inputs. Since the op-amps draw no (or negligible) current from their inputs, we can save power in our design by selecting resistors of some $\text{k}\Omega$ e.g. $R = 10 \text{ k}\Omega$



3.

(a) $2^6 = 64$, therefore we need 6 binary bits to represent the digital numbers 0 to 63, inclusive.

(b) 6 – one switch and resistor per bit.

(c) We need to add 3 new resistors and switches to represent the lowest 3 bits of our 6 bit number. Horowitz & Hill seem to be using resistor numbers from the E24 scale of standard values, so the next resistances closest to the power-of-two series that we want are 82k, 160k, 330k. See this [link](#) for the odd history of resistor values...

(d) For a digital input of zero, all switches will be in the 'down' position and $V_- = V_{out} = 0$. For a digital input of '1' only the LSB switch will be 'up' hence we have

Instrumentation

$$\frac{1}{330} = -\frac{V_{out}}{49.9}$$

$$V_{out} = -0.151 \text{ V.}$$

For a digital input of b111111, we have all switches in the 'up' position and we need to sum all of the currents which flow into the virtual ground:

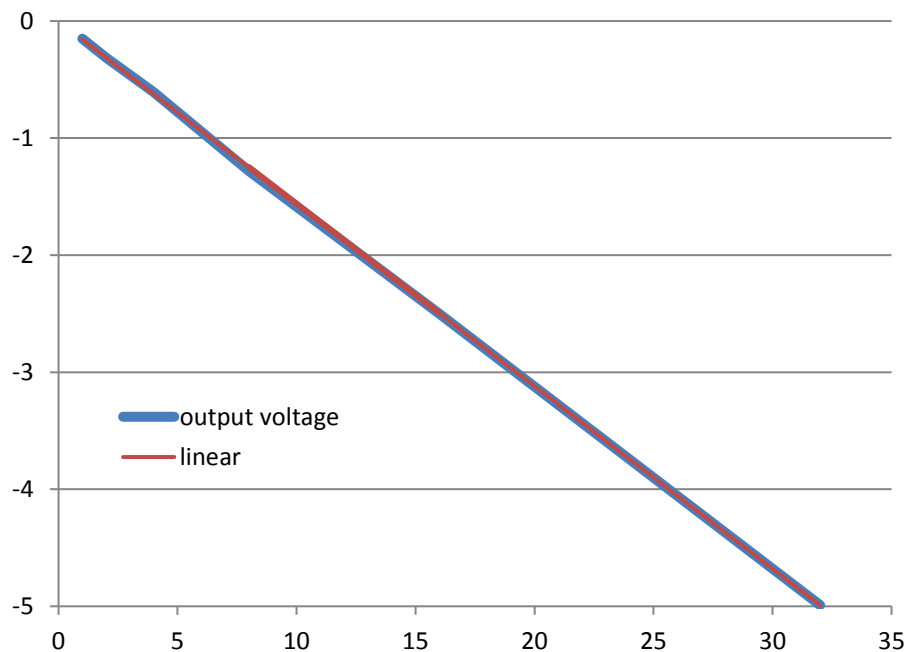
$$\frac{1}{10} + \frac{1}{20} + \frac{1}{39} + \frac{1}{82} + \frac{1}{160} + \frac{1}{330} = -\frac{V_{out}}{49.9}$$

$$V_{out} = -9.84 \text{ V}$$

(e) The table gives the output voltage for the linear case (ideal resistors in a power of 2 series) and for the case using 'real' values:

Input	Output (real resistors)	Output (linear)
1	-0.151	-0.156
2	-0.312	-0.312
4	-0.609	-0.624
8	-1.28	-1.25
16	-2.50	-2.50
32	-4.99	-4.99

From these numbers and the plot below, we see that the device is slightly non-linear due to the sequence of resistor values.



(f) Let us choose to set the MSB to '1' so we have a digital input of d32 and want an output voltage of -3.2 V

$$-\frac{3.2}{R_f} = -\frac{1}{10}$$

$$R_f = 32 \text{ k}\Omega$$

As mentioned, the output will not be *exactly* scaled by 0.1 due to the inherent non-linearity of the DAC.